

Harry Nguyen

+1 470-667-9000 | harrynguyenswe@gmail.com | linkedin.com/in/harrynguyen26 | github.com/HarryNguyen2662 | timoto.ai

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

WORK EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **Python-compatible C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99, 10K+ req/sec** — selected Protobuf for schema evolution and cross-language API compatibility
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions in Chrome stable channel
- Architected event-driven **TypeScript/React** system with observer pattern across **25K+ lines** of Chromium at **95% test coverage**; design documents adopted into production branch by senior engineers
- Led code reviews and delivered extensible, well-tested system designs with **95% test coverage** standard adopted across team

CoderPush | Software Engineer Intern

Sep 2023 – Dec 2023

- Designed idempotent **payment APIs** with Redis distributed caching (**85% cache hit rate**) and exponential backoff — enforced **exactly-once semantics** preventing duplicate transactions under network partitions at **9,000+ req/sec**
- Diagnosed DynamoDB hot-partition failure under sustained high-concurrency load; redesigned partition key strategy — **30% read throughput improvement**, system stability restored under peak traffic

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on **AWS** — selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

PERSONAL PROJECTS

TiMoto AI – Backend API & Search Platform | *Python, Django, gRPC, AWS, K8s, Terraform*timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end **Python/Django REST** backend with gRPC inter-service layer, PostgreSQL, and multi-AZ **AWS** infrastructure serving discovery and ranking workflows
- Resolved **production deadlock under concurrent gRPC calls** — traced root cause to circular resource acquisition; redesigned call sequencing — **100% API success rate** at **sub-50ms p99** under concurrent load
- Architected multi-AZ ECS Fargate with **Kubernetes-compatible Terraform IaC** and circuit breaker pattern — **99.9% uptime**, **44% cost reduction** to \$40–60/month; auto-rollback on health check failure
- Integrated **vLLM** inference engine with PagedAttention for ranking and evaluation pipeline — zero OOM failures at production traffic through continuous batching and resource isolation

Pulumi — Open Source Contributor | *Go, TypeScript, IaC*

24.4K★ github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, C++, Java, JavaScript, SQL, Bash, Rust

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, MySQL, Redis, DynamoDB, MongoDB

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions, GCP, Azure

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Raft/Paxos, Fault tolerance, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, PyTorch, LangChain, Ray